HSC BIOLOGY CURRICULUM SYLLABUS

National Curriculum and Textbook Board (NCTB) Bangladesh

1. Marks Distribution & Examination Pattern

Each Biology paper carries 100 total marks divided into three distinct segments:

1. Creative Questions (CQ) / Theory: 50 Marks (Answer 5 questions out of 8, each worth 10 marks: 1+2+3+4).

2. Multiple Choice Questions (MCQ): 25 Marks (25 questions, 1 mark each, 25 minutes duration).

3. Practical Assessment: 25 Marks (Specimen dissection/identification, temporary slide preparation, experiment, oral viva, and practical notebook).

2. Biology 1st Paper (Botany) Chapters

Chapter 1: Cell and Its Structure (Plasma membrane, organelle ultrastructure, chromosome morphology, DNA/RNA replication).

Chapter 2: Cell Division (Amitosis, Mitosis stages, Meiosis I/II and genetic crossing over).

Chapter 3: Cell Chemistry (Carbohydrates, amino acids, lipids, proteins, and enzyme kinetics).

Chapter 4: Microorganisms (Virus structure like T2 and HIV, Bacteria classification, Malarial parasite Plasmodium life cycle).

Chapter 5: Algae and Fungi (Spirogyra, Agaricus, Phytophthora, lichen symbiosis).

Chapter 6: Bryophyta and Pteridophyta (Riccia morphology, Pteris life cycle and alternation of generations).

Chapter 7: Gymnosperms and Angiosperms (Cycas morphology, floral formula, floral diagram of Poaceae and Malvaceae).

Chapter 8: Tissue and Tissue System (Meristematic tissue, vascular bundle arrangements, dicot and monocot stem/root anatomy).

Chapter 9: Plant Physiology (Water absorption, transpiration, photosynthesis C3/C4, cellular respiration, mineral nutrition).

Chapter 10: Plant Reproduction (Microsporogenesis, megasporogenesis, double fertilization, apomixis).

Chapter 11: Biotechnology (Tissue culture, recombinant DNA technology, PCR, genetic engineering applications).

Chapter 12: Environment, Distribution and Conservation (Biomes, Sundarbans mangrove ecosystem, in-situ and ex-situ conservation).

3. Biology 2nd Paper (Zoology) Chapters

Chapter 1: Animal Diversity and Classification (Symmetry, coelom, germ layers, Non-Chordata and Chordata phyla characteristics).

Chapter 2: Introduction to Animals (Hydra morphology and nematocysts, Periplaneta americana anatomy, Labeo rohita).

Chapter 3: Digestion and Absorption (Digestive tract, liver, pancreas, enzymatic breakdown of carbohydrates, proteins, lipids).

Chapter 4: Blood Circulation (Heart structure, cardiac cycle, ECG, coronary circulation, baroreceptors, cardiovascular diseases).

Chapter 5: Respiration and Breathing (Alveoli histology, oxygen-hemoglobin dissociation curve, carbon dioxide transport, respiratory volumes).

Chapter 6: Waste and Excretion (Nephron histology, ultrafiltration in glomerulus, counter-current mechanism, osmoregulation).

Chapter 7: Locomotion and Movement (Human skeletal framework, bone histology, muscle sliding filament theory, joints).

Chapter 8: Coordination and Control (Brain regions, cranial nerves, endocrine glands and hormonal actions, eye and ear anatomy).

Chapter 9: Continuation of Human Life (Gametogenesis, menstrual cycle, fertilization, embryonic development, placenta).

Chapter 10: Defence of the Human Body (Innate immunity, humoral and cell-mediated immunity, antibodies, vaccination).

Chapter 11: Genetics and Evolution (Mendelian genetics, epistasis, sex-linked disorders, Darwinism and neo-Darwinian evolution).

Chapter 12: Animal Behaviour (Innate vs learned behaviour, circadian rhythms, altruism in honeybees, migration in birds).